Analytical Chemistry Cumulative Exam
April 26, 2003
(Electrochemistry, each question is 20 points)

1. Describe a model of the double-layer region of an electrode in solution. Name the different layers in this region and explain what species are present in each layer and approximately how large each layer is.

2. In an amperometric electrochemical experiment where a redox species is in solution and oxidizing or reducing at the electrode, what are at least three factors that may cause limitations to the magnitude of the current?

3. Comment on whether microelectrodes or macroelectrodes have an advantage (if any) in carrying out electrochemistry of redox species under the following situations. Explain why.
   a. highly resistive solutions
   b. very short time scales for the experiment (fast electrochemical experiments)

4. A new organometallic molecule is synthesized. It is electroactive. You determine how many electrons are transferred per molecule by using a thin electrochemical cell, which has a known volume of 0.188 cm^3 (100 μm gap x 2.5 cm wide x 7.5 cm long). Integration of the peak under the CV response (very slow time scales) is 3.63×10^{-2} Coulombs for a concentration of 1×10^{-3} M. What is n (the number of moles of electrons per mole of molecules)? Note that the Faraday constant is 96485 Coulombs/mol of electrons.

5. A spoon with an area of 4 cm^2 is plated with a layer of an average thickness of 37.7 mils of silver from a Ag^+ solution (atomic weight of Ag = 107.8682 g/mol, density of Ag = 10.5 g/cm^3, 1 mil = 0.00254 cm). What current needs to be applied to achieve this thickness in 2 hours if the current efficiency is 80%? Note that the Faraday constant is 96485 Coulombs/mol of electrons.